

Automated Motor Rodage Tool

Industrial tooling for controlled and repeatable motor operation.

SYSTEM OVERVIEW

MOTOR CONTROL

Arduino Mega / L298N / Motor

ENGINEERING PROOF

Arduino Mega

Motor Control

Proteus

SolidWorks

Developed in an industrial engineering internship environment at Safran Electronics & Defense.

From manual operation to controlled tooling

How can a repetitive motor operation be transformed into a controlled and repeatable industrial process?

OBJECTIVE

Develop a practical tooling concept able to control motor operation through a defined sequence while considering operator interaction, repeatability and integration constraints.

INDUSTRIAL ENGINEERING / METHODS / EMBEDDED CONTROL

Project scope

- Controlled forward and reverse motor operation.
- Defined timing between operating phases.
- Operator-oriented control interaction.
- Arduino Mega based control architecture.
- Motor-driver interface using an L298N.
- Mechanical design and prototyping using SolidWorks.
- Functional simulation using Proteus.

Project at a glance

CONTEXT	Industrial engineering internship
CONTROL	Arduino Mega
DRIVER	L298N
MECHANICAL DESIGN	SolidWorks
SIMULATION	Proteus

Control architecture

The system is structured around a simple control chain: operator interaction, embedded logic, motor-driver interface and motor actuation.



Control layer

The Arduino Mega acts as the embedded control layer. It receives the operator input and determines the motor operation sequence.

Motor interface

The L298N provides the interface between the controller and the motor, allowing the direction of rotation to be controlled.

Operator interaction

The tooling is designed around a simple operator-oriented interaction model, reducing the need for repeated manual motor manipulation.

Repeatability through control.

The engineering objective is not simply to make the motor turn, but to convert an operator-dependent repetitive operation into a defined and reproducible sequence.

Mechanical integration

The tooling combines mechanical design, embedded control and electrical interfacing into a practical prototype concept.

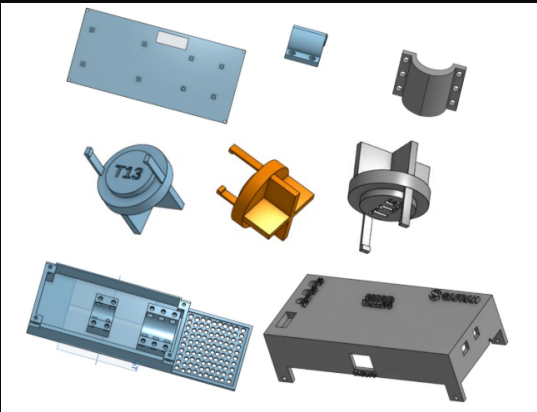


FIG. 01 SolidWorks — main tooling design

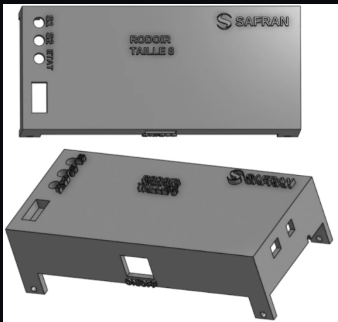


FIG. 02

SolidWorks — detail

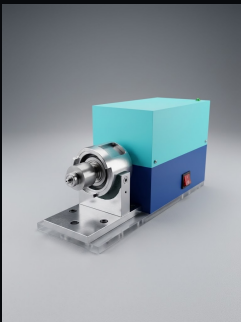


FIG. 03

Prototype / project view

MECHANICAL + ELECTRICAL + CONTROL

The tooling concept brings together the physical structure, motor interface and embedded control layer.

The control logic

A compact Arduino implementation translates the operating sequence into explicit motor commands, direction changes and timing intervals.

ARDUINO / MOTOR CONTROL COREC++ / EMBEDDED CONTROL

```
1 // Motor direction control
2 digitalWrite(IN1, HIGH);
3 digitalWrite(IN2, LOW);
4 analogWrite(ENA, motorSpeed);
5
6 delay(forwardTime);
7
8 // Stop motor
9 digitalWrite(IN1, LOW);
10 digitalWrite(IN2, LOW);
11
12 delay(pauseTime);
13
14 // Reverse direction
15 digitalWrite(IN1, LOW);
16 digitalWrite(IN2, HIGH);
17 analogWrite(ENA, motorSpeed);
18
19 delay(reverseTime);
```

WHAT THE CODE DEMONSTRATES
Direction control through the motor-driver inputs. **PWM control** through the enable signal. **Timed sequencing** through defined operating intervals.
REPRESENTATIVE CONTROL LOGIC EXCERPTS - FULL SOURCE CODE MAINTAINED SEPARATELY

Engineering logic

INPUT	Operator interaction initiates the operating sequence.
FORWARD	The controller commands one motor direction for a defined period.
PAUSE	The motor is stopped before the direction change.
REVERSE	The controller switches the driver inputs to reverse the motor direction.

From logic to verification

The control concept was translated into a programmable sequence and examined through simulation before practical integration.

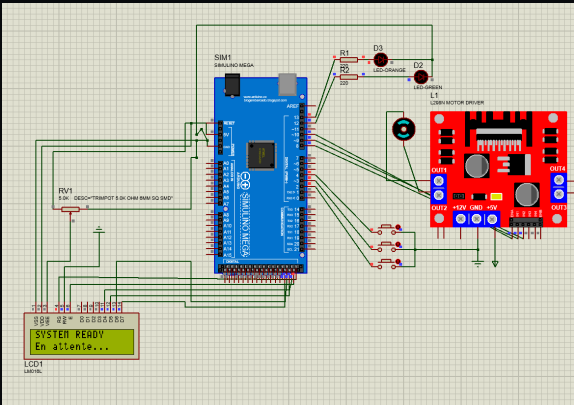


FIG. 04 Proteus simulation

- 01
- Input**
Operator interaction initiates the operating sequence.
- 02
- Control**
The Arduino Mega executes the defined motor-control logic.
- 03
- Actuation**
The L298N interfaces the controller with the motor.
- 04
- Sequence**
Forward operation, timed pause and reverse operation form the defined operating cycle.

VALIDATION APPROACH

Simulation provides a controlled environment for checking the logic and interaction between the controller, motor driver and motor before relying on the physical implementation.

What this project demonstrates

01

Industrial Automation

Translating a repetitive industrial task into a structured automated sequence.

02

Embedded Control

Using a microcontroller to coordinate operator input and motor-control logic.

03

Motor Control

Controlling motor direction and sequencing through an embedded motor-driver interface.

04

Mechanical Integration

Connecting mechanical design with the electrical and embedded control architecture.

05

Simulation

Using Proteus as a verification step for the control concept and system interaction.

06

Operator-Oriented Design

Designing the control approach around a practical industrial operator workflow.

ENGINEERING SNAPSHOT

Arduino Mega · L298N · Motor Control · SolidWorks · Proteus